



E.A. NUWINA NIDUWARA ELABADAARACHCHI

Mechanical Engineering Undergraduate

Fueled by passion and purpose, I am a Mechanical and Manufacturing Engineer with a strong foundation in **Automation, Design, Maintenance, Production, and Management**. I thrive in dynamic environments and embrace challenges as opportunities to grow. As a self-driven learner, I adapt swiftly to emerging technologies and bring a problem-solving mindset to every project I take on.

☎ [\(+94\) 773755486](tel:+94773755486) ✉ enn24elabadaarachchinuwina@gmail.com 🌐 [Nuwina Niduwara](#) 📍 Badulla, Sri Lanka

EDUCATION

Badulla Central College

- G.C.E. Advanced Level (Physical Science): 2A's and 1B
- G.C.E. Ordinary Level: - 9A's

University of Ruhuna Faculty of Engineering

- B.Sc. (Hons) Engineering 3rd year Undergraduate Department of Mechanical and Manufacturing Engineering
- Relevant Coursework: Computer Aided Designing and Manufacturing, Mechatronics, Advanced Controls, Electric Machines, Maintenance management, Automobile, Material manufacturing, Fluid Dynamics, Mechanics of Machines, Mechanics of Materials, Project Management.

WORKING EXPERIENCE

Production/Maintenance Engineering Intern

Nestlé Lanka Limited, Kurunegala

Aug 2025 – Oct 2025

- Gained hands on experience in production engineering and manufacturing operations.
- Assisted with Preventive Maintenance (PM) activities.
- **Gasket Management System Verification Nestlé Lanka PLC**
Verified and audited the Milk Reception, Evaporator, Dry Mixing, and Vending Mix gasket management systems by identifying mismatched gasket records, correcting discrepancies, updating equipment data, and ensuring accurate documentation to improve maintenance reliability, inventory accuracy, and preventive maintenance efficiency.
- Interpreted Piping and Instrumentation Diagrams (P&ID) for production processes.
- Conducted gasket mapping and supported maintenance activities in production areas.
- Developed practical knowledge of industrial equipment, process optimization, and workplace safety.

Samanthi Driving School (pvt) ltd

April 2024-present

- Marketing /social media coordinator ([Graphic design/adobe photoshop](#))

EMPLOYMENT

- **Member**

Institution of Mechanical Engineer (IMechE)

April 2025-present

IESL student member

April 2025-present

- **Marketing and publicity Co lead**

Xbotix 2026 (IEEE RUHUNA CHAPTER)

July 2026-present

- **Committee member**

Innovation club of Ruhuna (ICORE)

April 2025-present

- **Organizing committee marketing team member**

MEXCELLANCE 2025 /mechanical and manufacturing engineering student society (MMESS)

April 2025-present

- **Organizing committee member Design team**
Rextro 2025 Exhibition 25th anniversary faculty of engineering university of ruhuna **April 2025-present**
- **Volunteer**
Mehewara **Jan 2023-present**
- **Volunteer**
Sasnaka sansada Foundation **Jan 2022-Jan 2023**

SKILLS

- Technical Skills: AutoCAD | SolidWorks | Adobe Photoshop | Arduino | Porteous | Figma | inscape | FluidSim | MatLab & Simulink | FEA | Exell |
- Soft Skills: Leadership | Problem solving | Communication | Flexibility | Punctuality | Teaching Skills | Collaborative | Presentation Skills

PROJECTS/AND

- **Design of Solar Powered Water Pump System**
Contributed to a group academic project focused on designing a solar powered water pump system using SolidWorks. The project involved developing a detailed 3D prototype and a comprehensive project proposal, addressing technical specifications, sustainability aspects, and practical feasibility. Additionally, we explored the marketing potential of the product, including target users, cost analysis, and value proposition for agricultural applications.
- **Design and Development of an Articulated Arm for Surface Grinding of Distance Blocks (Ongoing)**
Designing and developing a gas strut based articulated balancing arm to support an angle grinder during the surface grinding of railway distance blocks at the **Department of Government Factory, Kolonnawa**. The project focuses on improving operator ergonomics by reducing muscle fatigue and hand-arm vibration while enhancing grinding efficiency. Responsibilities include field data collection, mechanism selection, CAD modelling, material and component selection, load and stress analysis, prototype development, and engineering validation.
- **Portable Tube Well Drilling Machine (Engineering Design Project)**
Designed a portable trailer-mounted tube well drilling machine capable of drilling up to 100 m with a 100–200 mm bore diameter for domestic and agricultural groundwater applications. Performed structural design, component selection, mechanical calculations, and CAD modeling while focusing on portability, safety, manufacturability, and the use of locally available industrial components.
- **Autonomous Maze-Solving Robot | ESP32 Mechatronics Project**
Designed and developed an autonomous line-following robot using an ESP32 microcontroller to solve maze environments. Integrated IR sensors, DC gear motors, and embedded control algorithms for line tracking, junction detection, path optimization, and Bluetooth-based monitoring, while contributing to the mechanical design and system integration.
- **3D Model of 4-Stroke Engine**
Designed a 3D model of a four-stroke internal combustion engine to demonstrate piston, crankshaft, and valve movements. Used Solidworks software to simulate the complete engine cycle for educational visualization.
- **4-Bit Adder and Subtractor using Proteus**
Built and simulated a 4-bit binary adder-subtractor circuit using basic logic gates in Proteus Design Suite. Verified digital logic operations and improved circuit design skills through simulation.

LEADERSHIP & ACTIVITIES

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|--|-------------------|
| • Captain of Faculty Basketball Team - Present | Jan 2025-present |
| • Vice Captain University basketball team -present | |
| Ruhuna basketball | Jan 2025-present |
| • U17 and U19 Basketball team captain | |
| Badulla Central College, Badulla district, uva province ,Sri Lanka | Jan 2019-Jan 2022 |
| • Quiz club member/sirasa pentathlon team member | |
| Badulla Central College, Badulla district, uva province ,Sri Lanka | Jan 2019-Jan 2020 |

AFFILIATIONS

- Sri Lanka University Games BASKETBALL 2nd Runner Up 2023 (SLUG)
- University Color Holder Basketball 2023
- National Sport Meet Quiz Competition Badulla District Champions 2020
- Sirasa Pentathlon Quarter Finals 2019

REFERENCE

- Dr Ruwan Gallage Senior Lecture Department of Mechanical and manufacturing engineering
University of Ruhuna. |+94912245765/6/7 | rgallage@mme.ruh.ac.lk